The documentation and process conversion measures necessary to comply with this revision shall be completed by 18 October 1997.

INCH POUND

MIL-PRF-19500/255J
18 April 1997
SUPERSEDING
MIL-S-19500/255H
25 JULY 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN,
SILICON, SWITCHING,
TYPES 2N2221A, 2N2221AL, 2N2222A, 2N2222AL, 2N2221AUA
2N2222AUA, 2N2221AUB, AND 2N2222AUB
JAN, JANTX, JANTXV, JANTXVD, JANTXVH, JANTXVM, JANTXVR,
JANS, JANSD, JANSH, JANSM, JANSR, JANHC, AND JANKC

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Provisions for radiation hardness assurance (RHA) to four radiation levels is provided for JANTXV and JANS product assurance levels. RHA level designators "M", "D", "R", and "H" are appended to the device prefix to identify devices which have passed RHA requirements.

1.2 Physical dimensions. See figures 1 and 2 (similar to T0-18), figures 3 and 4 (surface mount case outlines UA and UB), and figures 5, 6, and 7(JANHC and JANKC).

1.3 Maximum ratings.

Types	P_T 1/, 4/ T_A = +25°C	P_T 2/,3/ T_C = +25°C	I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
	W	W	mA dc	V dc	V dc	V dc	°C
2N2221A,L,UA,UB	0.5 1/	1.8 2/ 3/	800	75	50	6	-65 to +200
2N2222A,L,UA,UB	0.5 1/	1.8 2/ 3/	800	75	50	6	-65 to +200

1/ Derate linearly 2.85 mW/°C for TA > +25°C.
2/ Derate linearly 10.3 mW/°C for TC > +25°C (Nonsurface mount devices).
3/ For UA and UB surface mount case outlines, P_T = 1.16 W; derate linearly 6.6 mW/°C for T_C > +25°C.
4/ For UB surface mount case outlines, P_T = 300 mW; derate linearly 1.71 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

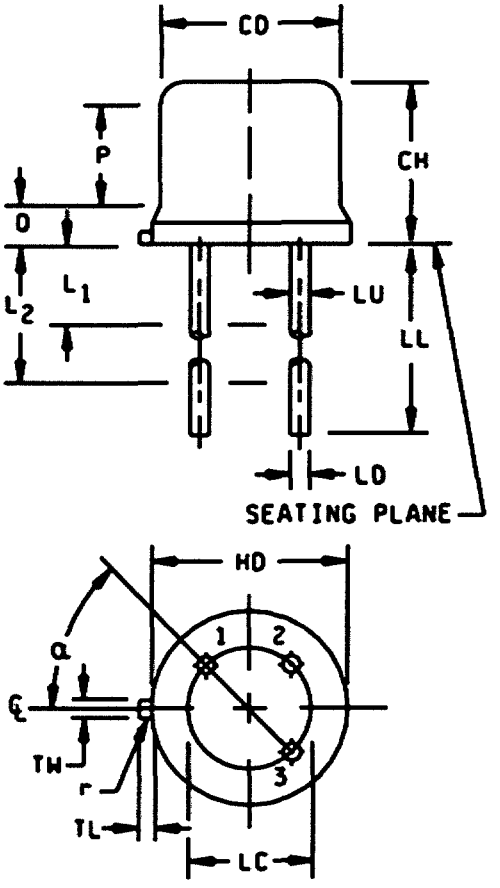
AMSC N/A

FSC 5961

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Symbol	Inches		Millimeters		Note
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	.8255	.8509	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8,13
LU	.016	.019	0.41	0.48	7,8
L_1	---	.050	---	1.27	7,8
L_2	.250	---	6.35	---	7,8
Q	---	.030	---	0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r	---	.010	---	0.25	10
α	45° TP		45° TP		6

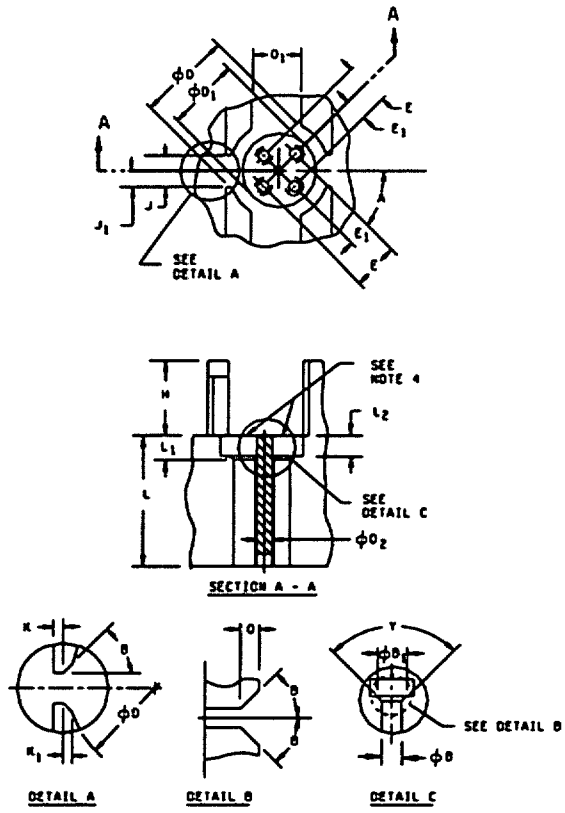


NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.
13. For L suffix devices, dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max.

FIGURE 1. Physical dimensions (similar to TO-18).

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Ltr	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
ϕB	.8255	.8509	.0325	.0335	5
ϕB_1	1.09 Nom		.040 Nom		5
ϕD	5.867	5.880	.2310	.2315	
ϕD_1	4.04	4.09	.159	.161	
ϕD_2	1.02 Nom		.040 Nom		6
E	2.527	2.553	.0995	.1005	
E_1	1.257	1.283	.0495	.0505	
H	3.68	3.94	.145	.155	
J	1.194	1.207	.0470	.0475	
J_1	.597	.622	.0235	.0245	
K	.229	.279	.009	.011	
K_1	.127 Nom		.005 Nom		
L	9.45	9.60	.372	.378	
L_1	1.37	1.40	.054	.055	
L_2	1.09 Nom		.043 Nom		
Q	1.02 Nom		.040 Nom		
Q_1	3.12	3.23	.123	.127	
α	---	---	44.90°	45.10°	
B	---		45° Nom		
Y	---		90° Nom		

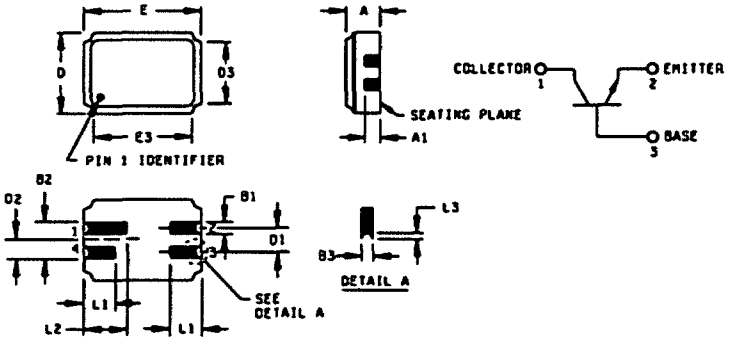


NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) ±.010 inch (0.254 mm) from the seating surface of the gauge. A force of 8 ±.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
4. These surfaces to be parallel and in same plane within ±.001 inch (0.025 mm).
5. Four holes.
6. Pressed in.

FIGURE 2. Gauge for lead and tab location.

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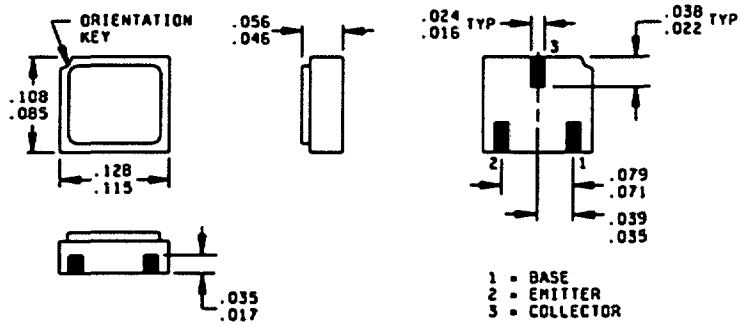


Dimensions					
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.061	.075	1.55	1.90	3
A_1	.029	.041	0.74	1.04	
B_1	.022	.028	0.56	0.71	
B_2	.075 REF		1.91 REF		
B_3	.006	.022	0.15	0.56	5
D	.145	.155	3.68	3.93	
D_1	.045	.055	1.14	1.39	
D_2	.0375 BSC		.952 BSC		
D_3	---	.155	---	3.93	
E	.215	.225	5.46	5.71	
E_3	---	.225	---	5.71	
L_1	.032	.048	0.81	1.22	
L_2	.072	.088	1.83	2.23	
L_3	.003	.007	0.08	0.18	5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension "A" controls the overall package thickness. When a window lid is used, dimension "A" must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions "B3" minimum and "L3" minimum and the appropriately castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension "B3" maximum and "L3" maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.

FIGURE 3. Physical dimensions, surface mount (UA version).

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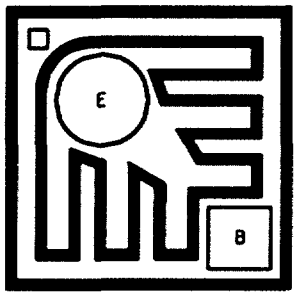


Inches	mm	Inches	mm
.016	0.41	.056	1.42
.017	0.43	.071	1.81
.022	0.56	.079	2.01
.024	0.61	.085	2.41
.035	0.89	.108	2.74
.038	0.96	.115	2.82
.039	0.99	.128	3.25
.046	0.97		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Physical dimensions, surface mount (UB version).

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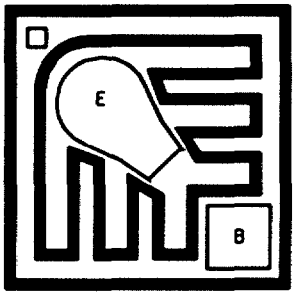
Physical characteristics:
A-version

Die size:	0.020 by 0.020 inches square
Die thickness:	0.0120 ±0.0018 inches
Base:	0.004 x 0.004 inches
Emitter:	0.004 inches diameter
Back metal:	Gold 5 KÅ ±1 KÅ
Top metal:	Aluminum 12,000 KÅ, ±2 KÅ
Back side:	Collector
Glassivation:	Si_3O_2, 8 kÅ nominal.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified tolerance is ±0.00100 inch (0.00254 mm).

FIGURE 5. JANHC and JANKC (A-version) die dimensions.

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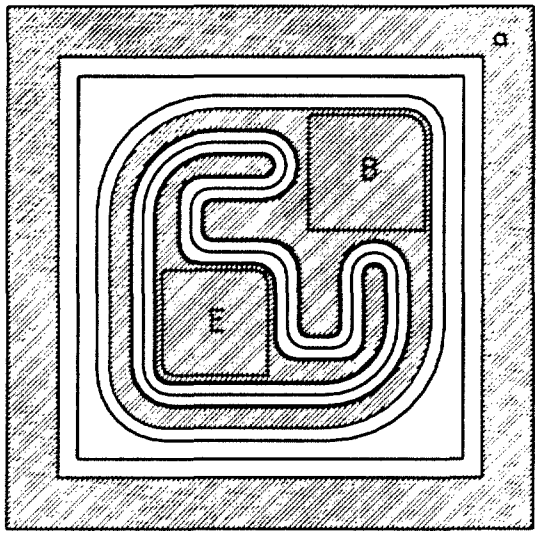
Physical characteristics:
B-version

Die size: 0.020 by 0.020 inches square, ±0.001
Die thickness: 0.0090 ±0.0015 inches
Base: 0.005 square
Emitter: 0.006 by .004 inches
Back metal: Gold 4.5 KÅ ±1 KÅ
Top metal: Aluminum 15,000 KÅ, ±2 KÅ
Back side: Collector
Glassivation: Si_3N_4, 2 kÅ nominal.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified tolerance is ±0.00100 inch (0.00254 mm).

FIGURE 6. JANHC and JANKC (B-version) die dimensions.

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Physical characteristics:
C-version

Die size:	0.0205 by 0.0205 inches, ±0.001
Base:	0.0045 square
Emitter:	0.0041 by .0041 inches
Back metal:	Gold
Top metal:	Aluminum
Back side:	Collector

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified tolerance is ±0.00100 inch (0.00254 mm).

FIGURE 7. JANHC and JANKC (C-version, 2N2222A only) die dimensions.

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1.4 Primary electrical characteristics at T_A = +25°C.

	h_{FE} at V_{CE} = 10 V dc									
	h_{FE1} I_C = 0.1 mA dc		h_{FE2} I_C = 1.0 mA dc		h_{FE3} I_C = 10 mA dc		h_{FE4} 1/ I_C = 150 mA dc		h_{FE5} 1/ I_C = 500 mA dc	
	L, UA, UB		L, UA, UB		L, UA, UB		L, UA, UB		L, UA, UB	
	2N2221A	2N2222A	2N2221A	2N2222A	2N2221A	2N2222A	2N2221A	2N2222A	2N2221A	2N2222A
Min	30	50	35	75	40	100	40	100	20	30
Max			150	325			120	300		

Types	Limit	$\|h_{fe}\|$ f = 100 MHz V_{CE} = 20 V dc I_C = 20 mA dc	C_{obo} 100 kHz ≤ f ≤ 1 MHz V_{CB} = 10 V dc I_E = 0	Switching (saturated)	
				t_{on} See fig. 8	t_{off} See fig. 9
			pF	ns	ns
2N2221A, 2N2222A L, UA, UB	Min	2.5			
	Max		8	35	300

Types	Limits	$V_{CE(sat)1}$ 1/ I_C = 150 mA dc I_B = 15 mA dc	$V_{CE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)1}$ 1/ I_C = 150 mA dc I_B = 15 mA dc	$V_{BE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc
		V dc	V dc	V dc	V dc
2N2221A, 2N2222A L, UA, UB	Min			0.6	
	Max	0.3	1.0	1.2	2.0

1/ Pulsed see 4.5.1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

UA, UB, - - - - - - - Surface mount case outlines (see figures 3 and 4).

3.3 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1, 2, 3, 4, 5, 6, and 7 herein. Epoxy die attach may be used when a moisture monitor plan has been submitted and approved by the qualifying activity.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "AUB" suffix can also be omitted.

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.

3.6 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
7	hermetic seal (optional)	1/
9	I_{CBO1}, h_{FE2}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO1}; I_{CES}; h_{FE2}; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ±15%	I_{CBO1}, h_{FE4}
12	See 4.3.2 240 hours minimum	See 4.3.2 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = ±15%	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15%

1/ Hermetic seal test shall be performed in either screen 7 or screen 14.

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, appendix H. As a minimum, die shall be 100 percent probed to insure the assembled chips will meet the requirements of group A, subgroup 2.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5;

JANS level (all device types) - - - - - - V_{CB} = 10 - 30 Vdc, P_T = 400 mW.
JANTX and JANTXV levels
2N2221A, L, UA; 2N2222A, L, UA - - - - - V_{CB} = 30 Vdc, P_T = 400 mW.
2N2221AUB, 2N2222AUB - - - - - V_{CB} = 30 Vdc, P_T = 200 mW.
NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current - - - - - - - - - - - 5 mA.
b. I_H forward heating current - - - - - - - - 200 mA (min).
c. t_H heating time - - - - - - - - - - - - - 25 - 30 ms.
d. t_{md} measurement delay time - - - - - - - - 60 µs max.
e. V_{CE} collector-emitter voltage - - - - - - - 10 V dc minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements shall be in accordance with the steps of table III herein as specified in the notes for table III.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc; P_T = 400 mW at T_A = room ambient per MIL-STD-750; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 V dc; P_T = 400 mW at T_A = 100°C for 96 hours, or T_A = +125°C ±25°C for 96 hours with P_T adjusted according to the chosen T_A to give an average T_J = +275°C.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	HTRB, test condition A, T_A = 150°C, V_{CB} = 50 V dc, 340 hours, n = 22 devices, c = 0.
2	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 V dc, P_T = 400 mW; P_T = 200 mW for UB; at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on the devices
	or	shall be permitted. n = 45 devices, Acc = 0
	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq$ +100°C; forced air cooling allowed on cooling cycle only. n = 45 devices, Acc = 0.
3	2037	Bond strength, n = 11 wires, c = 0.
4	1032	High-Temperature life (non-operating), T_A = +200°C n = 22, c = 0

1/ Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Samples shall be selected from one wafer lot every three months from a die fabrication line that the manufacturer chooses to "certify" for the die design used in products manufactured to this specification.

b. Must be chosen from an inspection lot that has been submitted to and passed group A conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B3 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.2.4 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die meeting the following requirements:

a. Manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

b. The die lot was manufactured (using start or completion dates, as the manufacturer selects to consistently use) within three months following the wafer lot used for sample selection in paragraph 4.4.2.3. Group B shall not be required more than once for any single wafer lot.

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4.4.3 Group C inspection, JANS. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 herein (JANS). See 4.4.3.2 for JAN, JANTX, and JANTXV group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with the steps of table III herein as specified in the notes for table III.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UA and UB devices.
C6	1026	conditions: 1,000 hours at V_{CB} = 10 V dc; P_T = 400 mW; P_T = 200 mW (for UB) at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on device shall be permitted.

4.4.3.2 Group C inspection, JAN, JANTX, and JANTXV. Group C inspection (JAN, JANTX, and JANTXV, package related tests) shall be as follows and shall include those package or case-related tests which are performed periodically. Group C test on each package family shall be performed on devices from each six months production of devices (based on inspection lot identification codes) for each assembly location and die attach method. Electrical measurements (end points) and delta requirements shall be in accordance with the steps of table III herein as specified in the notes for table III. The sample size for each of these subgroups is: 22 devices, c = 0. Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

Step	Method	Condition
1	2066	
2	1051	Test condition C.
3	2036	Test condition E; not applicable for UA and UB devices.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.3.4 Product acceptable for delivery. The product which may be accepted for delivery based a successful completion of group C testing shall be product of the particular package type with lot inspection identification codes of the 52 successive weeks beginning with the inspection lot identification code of the successful group C sample (i.e. the inspection lot identification codes accepted for delivery based on successful group C testing will include those sealed during the approximate 10 weeks during which the sample was being screened and tested plus those sealed in the 42 weeks following successful completion of group C testing making a total of 52 successive calendar weeks.) A manufacturer shall not accept product for delivery until after successful completion of the group C testing on product of the applicable package type with an inspection lot identification code less than 51 weeks prior to the inspection identification code of the product being considered for acceptance.

4.4.5 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II. These tests shall be performed as required in accordance with MIL-PRF-19500 and MIL-STD-750, method 1019 for total ionizing dose or method 1017 for neutron fluence as applicable.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

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TABLE I. Group A Inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071	45 devices, c = 0				
Solderability 1/ 2/	2026	15 leads, c = 0				
Resistance to solvents 2/ 3/ 4/	1022	15 devices, c = 0				
Subgroup 2						
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 10 uA dc	$V_{(BR)CBO}$	75		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 uA dc	$V_{(BR)EBO}$	6		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	50		V dc
Collector to emitter cutoff Current	3041	Bias condition C; V_{CE} = 50 V dc	I_{CES}		50	nA dc
Collector to base cutoff Current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}			
2N2221A, L, UA, UB				30		
2N2222A, L, UA, UB				50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}			
2N2221A, L, UA, UB				35	150	
2N2222A, L, UA, UB				75	325	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}			
2N2221A, L, UA, UB				40		
2N2222A, L, UA, UB				100		

See footnotes at end of table.

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TABLE I. Group A Inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N2221A, L, UA, UB				40	120	
2N2222A, L, UA, UB				100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed see 4.5.1	h_{FE5}			
2N2221A, L, UA, UB				20		
2N2222A, L, UA, UB				30		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N2221A, L, UA, UB					0.3	V dc
2N2222A, L, UA, UB						
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N2221A, L, UA, UB					1.0	V dc
2N2222A, L, UA, UB						
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mAdc; pulsed (see 4.5.1)	$V_{BE(sat)1}$			
2N2221A, L, UA, UB				0.6	1.2	Vdc
2N2222A, L, UA, UB						
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$			
2N2221A, L, UA, UB					2.0	V dc
2N2222A, L, UA, UB						
Subgroup 3						
High temperature operation		T_A = +150°C				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		10	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}			
2N2221A, L, UA, UB				15		
2N2222A, L, UA, UB				35		
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}			
2N2221A, L, UA, UB				30		
2N2222A, L, UA, UB				50		
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 20 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	2.5		
Open circuit Output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		25	pF
Saturated turn-on time		(See figure 8)	t_{on}		35	ns
Saturated turn-off time		(See figure 9)	t_{off}		300	ns
Subgroups 5 and 6						
Not required						
Subgroup 7 4/						
Decap internal visual (design verification)	2075	1 device, c = 0				

1/ For sampling plan see MIL-PRF-19500.
2/ Separate samples may be used.
3/ Not required for laser marked devices.
4/ Not required for JANS devices.

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TABLE II. Group D inspection

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Nuetron irradiation	1017					
Collector to base cutoff Current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		20	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}			
M2N2221A, D2N2222A, R2N2222A, H2N2222A				35 34 10		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}			
M2N2221A, D2N2222A, R2N2222A, H2N2222A				49 49 21	325	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}			
M2N2221A, D2N2222A, R2N2222A, H2N2222A				89 89 49		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc	h_{FE4}			
M2N2221A, D2N2222A, R2N2222A, H2N2222A				90 90 45	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc	h_{FE5}	27		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc;	$V_{CE(sat)1}$		.33	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc;	$V_{CE(sat)2}$		1.1	V dc

See footnotes at end of table.

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TABLE II. Group D inspection

Inspection	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Total dose irradiation	1019					
Collector to base cutoff Current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		20	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}			
M2N2221A, D2N2222A, R2N2222A, H2N2222A				45 34 10		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}			
M2N2221A, D2N2222A, R2N2222A, H2N2222A				67 60 21	325	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}			
M2N2221A, D2N2222A, R2N2222A, H2N2222A				90 90 50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc	h_{FE4}			
M2N2221A, D2N2222A, R2N2222A, H2N2222A				90 90 45		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc	h_{FE5}	27		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc;	$V_{CE(sat)1}$		.33	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc;	$V_{CE(sat)2}$		1.1	V dc

1/ Tests to be performed on all devices.
2/ For sampling plan, see MIL-PRF-19500.

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TABLE III. Groups A, B, and C electrical end-point inspection measurements. 1/, 2/, 3/, 4/, 5/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector-base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		10	nA dc
2.	Collector - base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		20	nA dc
3.	Emitter-base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO}		10	nA dc
4.	Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		.3	V dc
5.	Base-emitter voltage (saturated)	3066	Test condition A I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.2	V dc
6.	Forward current transfer ratio 2N2221A,L,UA,UB 2N2222A,L,UA,UB	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}	 35 75	 150 325	
7.	Collector-base cutoff current 2N2221A,L,UA,UB 2N2222A,L,UA,UB	3036	Bias condition D V_{CB} = 60 V dc	ΔI_{CBO} 1/	100% of initial value or 8 nA dc, whichever is greater.		
8.	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc; pulsed see 4.5.2	Δh_{FE2} 1/	±25% change from initial reading.		

1/ Devices which exceed the group A limits for this test shall not be accepted.

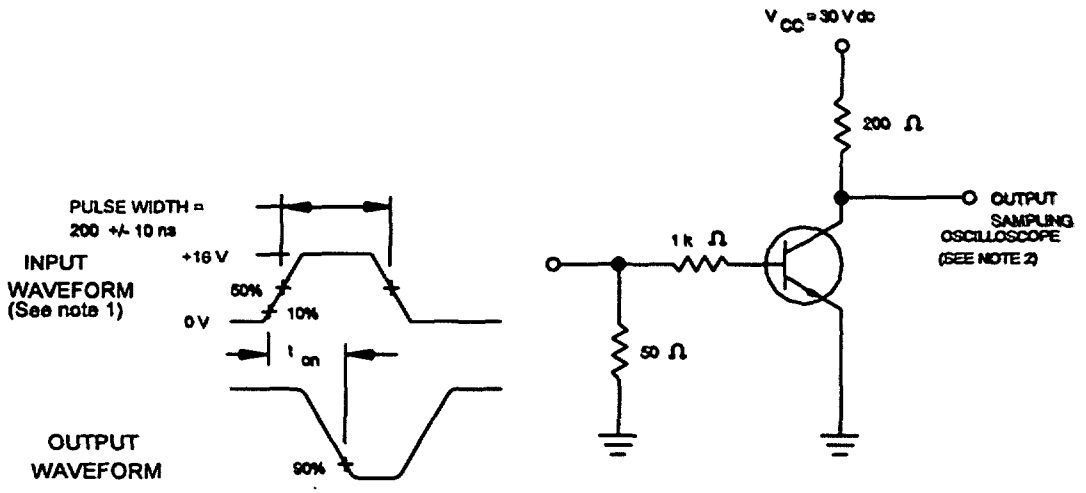
2/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
- a. Subgroup 3, see table III herein, steps 1, 3, 4, 5, and 6.
- b. Subgroups 4 and 5, see table III herein, steps 2, 3, 4, 5, and 8.

3/ The electrical measurements for group B, 4.4.2.2 herein(JAN, JANTX, and JANTXV,) are as follows:
- a. Steps 1, 2, and 4, of 4.4.2.2, see table III herein, steps 1, 6, 7, and 8.

4/ The electrical measurements for table VII (JANS) of MIL-PRF-19500 are as follows:
- a. Subgroup 2 and 3, see table III herein, steps 1, 2, 3, 4, 5, and 6 (for JANS).
- b. Subgroup 6, see table III herein, steps 2, 3, 4, 5, 6, 7, and 8 (for JANS).

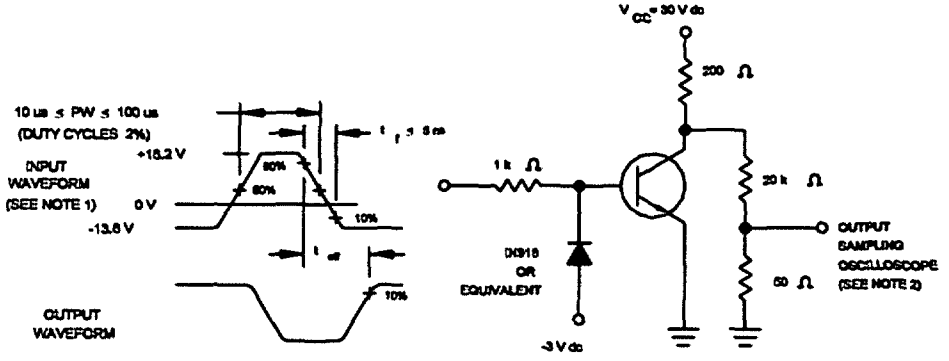
5/ The electrical measurements for group C, 4.4.3.2 herein (JAN, JANTX, and JANTXV) are as follows:
- a. Step 2 of 4.4.3.2, see table III herein, all steps.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 8. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 9. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead formation and finish may be specified (see 3.3.1).

b. Type designation and product assurance level.

c. For die acquisition, the JANHC or JANKC letter version shall be specified (see figure 5).

d. Surface mount designation if applicable.

e. Packaging requirements (see 5.1).

6.3 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification. This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N2221A) will be identified on the QPL.

Die ordering information 1/			
PIN	Manufacturer		
	07256	43611	32694
2N2221A 2N2222A	JANHCA2N2221A JANHCA2N2222A	JANHCA2N2221A JANHCB2N2222A	N/A JANHCA2N2222A

1/ For JANKC level, replace JANHC with JANKC.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1870)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/255J	2. DOCUMENT DATE (YYMMDD) 97/04/18

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING, TYPES 2N2221A, 2N2221AL, 2N2222A, 2N2222AL, 2N2221AUA , 2N2222AUA, 2N2221AUB, AND 2N2222AUB, JAN, JANTX, JANTXV, JANTXVD, JANTXVH, JANTXVM, JANTXVR, JANS, JANSD, JANSH, JANSM, JANSR, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0516
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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